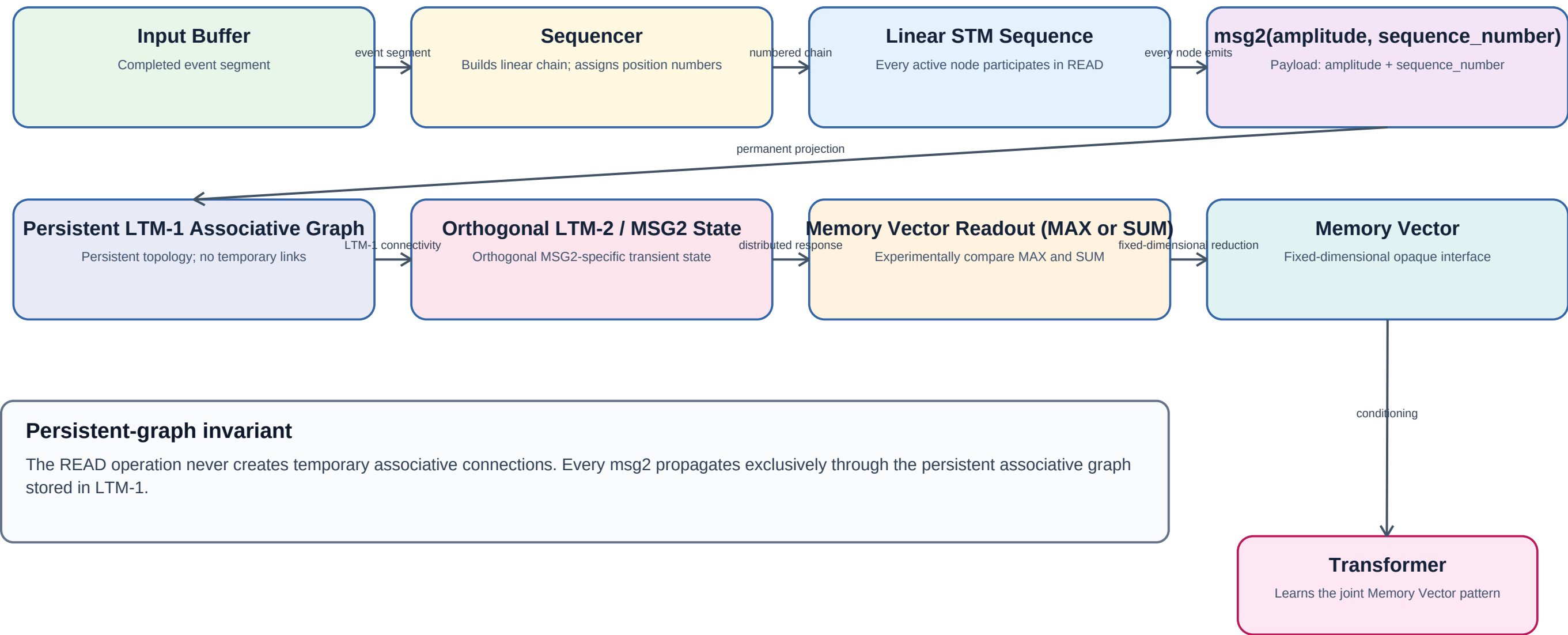


Cognitive Master Architecture Diagram

Release 0.3.48 - LTM-2 READ and Memory Vector generation



READ contract

1. Set operation mode to READ and reset Memory Vector output cells.
2. Sequencer numbers every node in the latest linear STM Sequence.
3. Every active node emits msg2(amplitude, sequence_number).
4. Permanent STM-to-LTM projections inject MSG2 into corresponding LTM inputs.
5. MSG2 propagates exclusively through the persistent LTM-1 associative graph.
6. LTM-2 adds no independent graph or propagation logic.
7. MAX or SUM reduces the distributed response to the fixed-dimensional Memory Vector.
8. Return Memory Vector and clear transient MSG2 state.

Information-preserving readout selection

Initial candidates: MAX and SUM.

Both use the same Memory Vector dimensionality, datasets, and numeric precision.

For deterministic READ, prefer the operator with higher non-degenerate output entropy.

For stochastic READ, raw entropy is insufficient; compare mutual information to exclude noise.

Coordinates are intentionally opaque and have no individual semantic interpretation.